1 Summary

Model (Run-name): model1 (test)

Iteration: 40

TAC: 154897.7828862593

Took: 4.746124982833862s

Total Time: 101.69859385490417s

Epsilons:

- ε_{LMTD}: 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area}: 1e-06
- ε_{Beta}: 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: `http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html`

2 Results

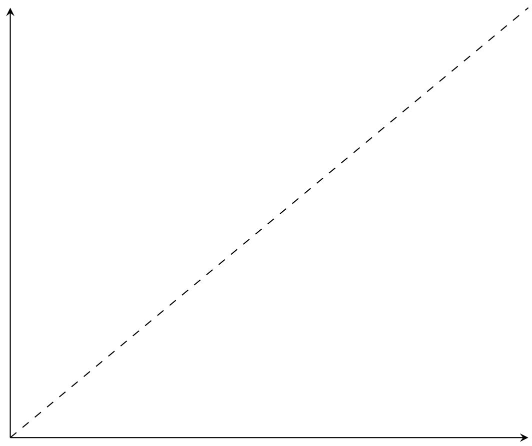
Variable	Value
$z_{1,1,1}$	1.0
$z_{1,2,2}$	1.0
$z_{2,1,2}$	1.0
$z_{cu,1}$	1.0
$z_{cu,2}$	1.0
$z_{hu,1}$	1.0
$A^{\beta}_{1,1,1}$	71.52154532022675
$A^{\beta}_{1,2,2}$	70.18046341267424
$A^{\beta}_{2,1,2}$	136.73553825185095
$A^{\beta}_{cu,1}$	4.701079869478363
$A^{\beta}_{cu,2}$	37.90741115359579
$A^{\beta}_{hu,1}$	13.341988022860814
$A_{1,1,1}$	71.52154532022675
$A_{1,2,2}$	70.18046341267424
$A_{2,1,2}$	136.73553825185095
$A_{cu,1}$	4.701079869478363
$A_{cu,2}$	37.90741115359579
$A_{hu,1}$	13.341988022860814
$dt_{1,1,1}$	32.97514314502193
$dt_{1,1,2}$	10.0
$dt_{1,2,2}$	81.07457056493419
$dt_{1,2,3}$	36.0745705649342
$dt_{2,1,2}$	18.92542943506581
$dt_{2,1,3}$	59.194072076299356
$dt_{cu,1}$	66.07457056493419
$dt_{cu,2}$	149.19407207629936
$dt_{hu,1}$	62.97514314502193
$f^{H}_{1,1,1}$	2.4616224798237787
$f^{H}_{1,2,2}$	10.0
$f^{H}_{2,1,2}$	10.982357083972783
$f^{C}_{1,1,1}$	14.493958168304086
$f^{C}_{1,2,2}$	13.0
$f^{C}_{2,1,2}$	10.207716218769342

Variable	Value
$RecLMTD_{1,1,1}$	0.05193302446912797
$RecLMTD_{1,2,2}$	0.017994990618634438
$RecLMTD_{2,1,2}$	0.028317110776040016
$RecLMTD_{cu,1}$	0.014701896798829944
$RecLMTD_{cu,2}$	0.009555372709395408
$RecLMTD_{hu,1}$	0.022487865159900346
$q_{1,1,1}$	689.2542943506658
$q_{1,2,2}$	1950.0
$q_{2,1,2}$	2416.1185584740133
$q_{cu,1}$	160.74570564934197
$q_{cu,2}$	1983.8814415259872
$q_{hu,1}$	494.627147175329
$t^{(H)}_{1,1}$	650.0
$t^{(H)}_{1,2}$	581.0745705649342
$t^{(H)}_{1,3}$	386.0745705649342
$t^{(H)}_{2,1}$	590.0
$t^{(H)}_{2,2}$	590.0
$t^{(H)}_{2,3}$	469.19407207629934
$t^{(C)}_{1,1}$	617.0248568549781
$t^{(C)}_{1,2}$	571.0745705649342
$t^{(C)}_{1,3}$	410.0
$t^{(C)}_{2,1}$	500.0
$t^{(C)}_{2,2}$	500.0
$t^{(C)}_{2,3}$	350.0
$t^{H}_{1,1,1}$	370.0
$t^{H}_{1,2,2}$	386.0745705649342
$t^{H}_{2,1,2}$	370.0
$t^{C}_{1,1,1}$	618.6104519264094
$t^{C}_{1,2,2}$	500.0
$t^{C}_{2,1,2}$	646.6953103605483
$b^{H,\text{in}}_{1,1,1}$	1600.054611885456
$b^{H,\text{in}}_{1,2,2}$	5810.745705649342
$b^{H,\text{in}}_{2,1,2}$	6479.590679543942

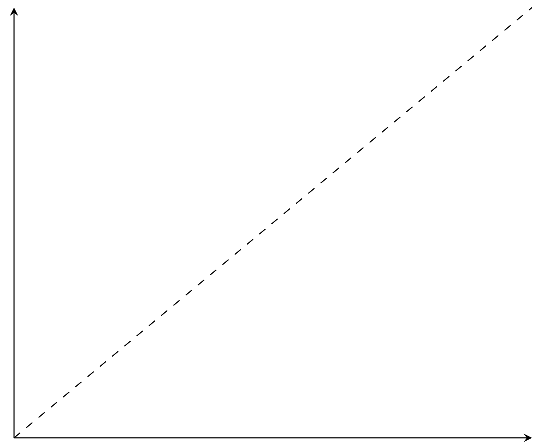
Variable	Value
$b^{H,\text{out}}_{1,1,1}$	910.8003175347981
$b^{H,\text{out}}_{1,2,2}$	3860.745705649342
$b^{H,\text{out}}_{2,1,2}$	4063.4721210699304
$b^{C,\text{in}}_{1,1,1}$	8277.204640905244
$b^{C,\text{in}}_{1,2,2}$	4550.0
$b^{C,\text{in}}_{2,1,2}$	4185.163649695431
$b^{C,\text{out}}_{1,1,1}$	8966.458935255901
$b^{C,\text{out}}_{1,2,2}$	6500.0
$b^{C,\text{out}}_{2,1,2}$	6601.282208169443

3 *RecLMTD* Tangent Points

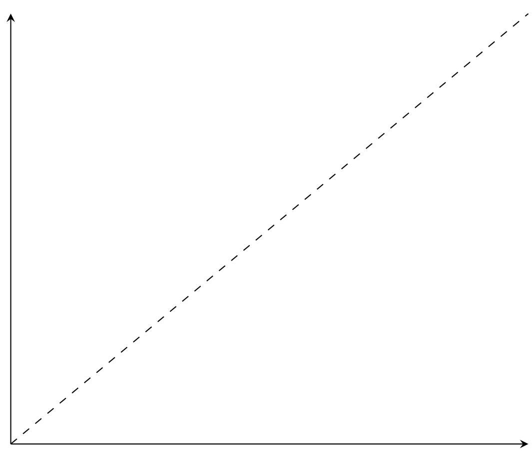
- $(1, 1, 1)$
  - Value: 0.051933
  - Error: 0.000000011990539
  - Relative error: 0.000000230884617
- $(1, 2, 2)$
  - Value: 0.017995
  - Error: 0.000000146154235
  - Relative error: 0.000008121874092
- $(2, 1, 2)$
  - Value: 0.028317
  - Error: 0.000000578566170
  - Relative error: 0.000020431263403
- $cu, 1$
  - Value: 0.014702
  - Error: 0.000000005691388
  - Relative error: 0.000000387119177
- $cu, 2$
  - Value: 0.009555
  - Error: 0.000000301013378
  - Relative error: 0.000031501010440
- $hu, 1$
  - Value: 0.022488
  - Error: 0.000000064926040
  - Relative error: 0.000002887150582



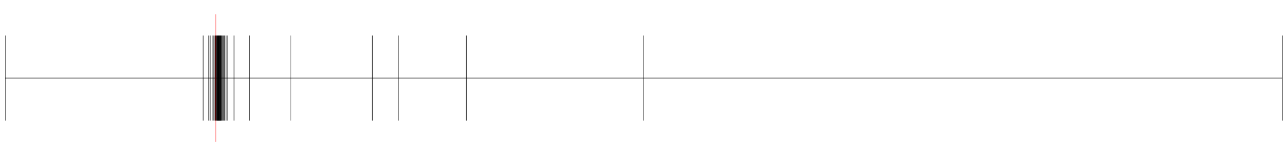
(a) *RecLMTD* - $(1, 1, 1)$



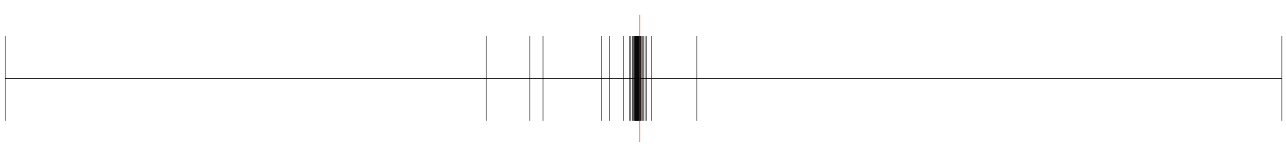
(b) *RecLMTD* - $(1, 2, 2)$



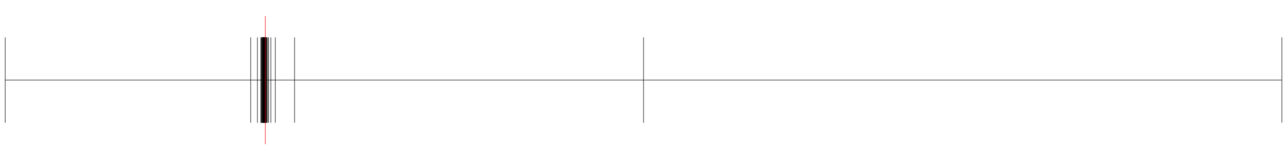
(c) *RecLMTD* - $(2, 1, 2)$



(a) *RecLMTD* - $cu, 1$



(b) *RecLMTD* - $cu, 2$



(a) *RecLMTD* - $hu, 1$

4 Balancing Breakpoints

4.1 $b^{\boldsymbol{H},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 1600.054611885456
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 5810.745705649342
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6479.590679543942
  - Absolute error: 0.0
  - Relative error: 0.0

4.2 $b^{\boldsymbol{H},\mathbf{out}}$

- $(1, 1, 1)$
  - Value: 910.8003175347981
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 3860.745705649342
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4063.4721210699304
  - Absolute error: 9.094947017729282e-13
  - Relative error: 2.2382206021717566e-16

4.3 $b^{\mathbf{C},\mathbf{in}}$

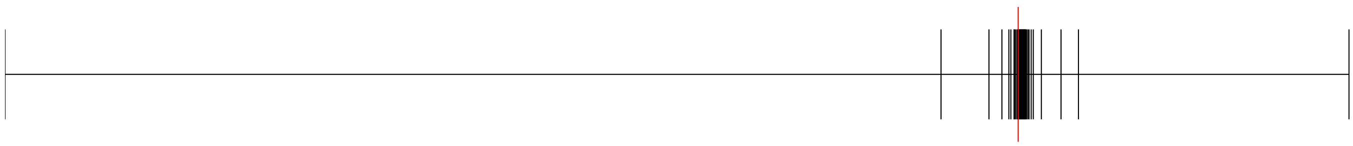
- $(1, 1, 1)$
  - Value: 8277.204640905244
  - Absolute error: 0.07370415486730053
  - Relative error: 8.904553453426096e-06
- $(1, 2, 2)$
  - Value: 4550.0
  - Absolute error: 0.0

  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4185.163649695431
  - Absolute error: 9.094947017729282e-13
  - Relative error: 2.1731401156538182e-16

4.4 $b^{\mathbf{C,out}}$

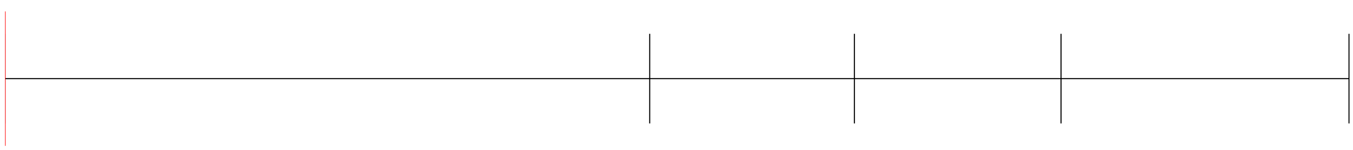
- $(1, 1, 1)$
  - Value: 8966.458935255901
  - Absolute error: 0.34492255883742473
  - Relative error: 3.846957091433079e-05
- $(1, 2, 2)$
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6601.282208169443
  - Absolute error: 9.094947017729282e-13
  - Relative error: 1.3777546135618617e-16

(a) $t^{(H)}$ - $(1,1)$



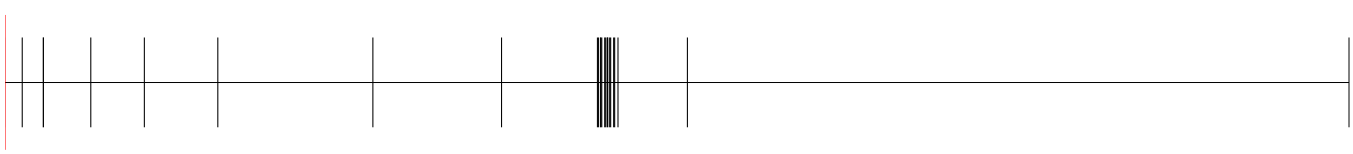
(a) $t^{(H)}$ - $(1,2)$

(a) $t^{(H)}$ - $(2,2)$



(a) $t^{(H)}$ - $(1,1,1)$

(a) $t^{(H)}$ - $(1,2,2)$



(a) $t^{(H)}$ - $(2,1,2)$

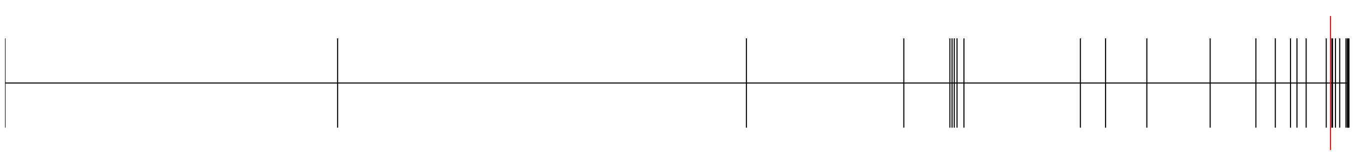
(a) $t^{(C)}$ - $(1,2)$

(a) $t^{(C)}$ - $(2,3)$

(a) $t^{(C)}$ - $(1,3)$

(a) $t^{(C)}$ - $(1,1,1)$

(a) $t^{(C)}$ - $(1,2,2)$



(a) $t^{(C)}$ - $(2,1,2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
  - Value: 71.521545
  - Error: 0.068574947701762
  - Relative error: 0.000957882839771
- $(1, 2, 2)$
  - Value: 70.180463
  - Error: 0.000000000000071
  - Relative error: 0.000000000000001
- $(2, 1, 2)$
  - Value: 136.735538
  - Error: 0.099455484858538
  - Relative error: 0.000726827854064

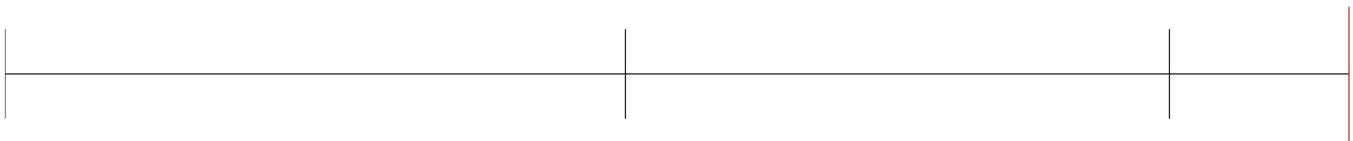
5.2 A_{CU}

- $cu, 1$
  - Value: 4.701080
  - Error: 0.025453681145080
  - Relative error: 0.005385274614569
- $cu, 2$
  - Value: 37.907411
  - Error: 0.006042016471092
  - Relative error: 0.000159363391248

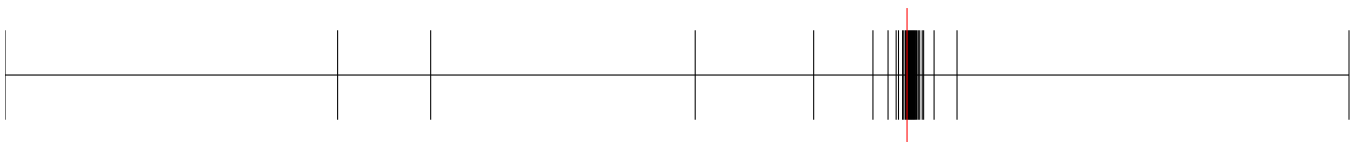
5.3 A_{HU}

- $hu, 1$
  - Value: 13.341988
  - Error: 0.005742285265164
  - Relative error: 0.000430206869079

(a) A^{β} - $(1, 1, 1)$

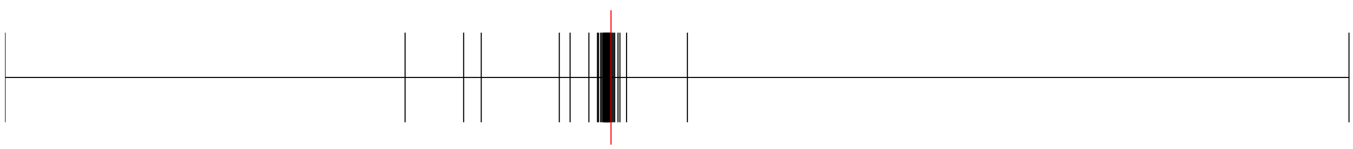


(a) A^{β} - $(1, 2, 2)$



(a) A^{β} - $(2, 1, 2)$

(a) A^{β} - $cu, 1$



(a) A^{β} - $cu, 2$

(a) A^{β} - $hu, 1$

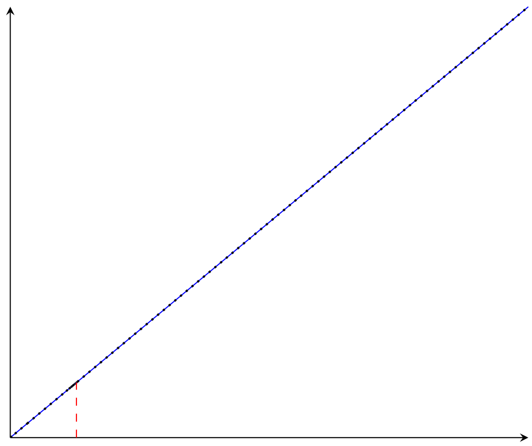
6 A^β Breakpoints

- $(1,1,1)$
  - q: 71.521545
  - $\hat{q}^\beta$: 71.521545
  - q^β: 71.521545
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.67982582933766, 70.76867866839005, 70.83181816751056, 70.8571173980136, 70.94547904612048, 70.95870595008942, 71.00935653192329, 71.03465240716494, 71.1226084153824, 71.18527282143235, 71.21162487917152, 71.22933662920877, 71.2616536557244, 71.30011482813455, 71.33750386361461, 71.3743618177527, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(1,2,2)$
  - q: 70.180463
  - $\hat{q}^\beta$: 70.180463
  - q^β: 70.180463
  - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.3940471550051, 68.48329505760876, 68.57240847861726, 68.66676327863097, 68.76047282726, 68.85281320938876, 68.94418284540035, 69.0314457108091, 69.13316279873226, 69.22655548799992, 69.3178258940278, 69.34502840170047, 69.41355730879985, 69.51025044804472, 69.60484346150977, 69.70052578983292, 69.93657924014246, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(2,1,2)$
  - q: 136.735538
  - $\hat{q}^\beta$: 136.735538
  - q^β: 136.735538
  - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, 137.63821724432228, 138.55705332155773, 138.86359281280824, 139.27702937706263, 139.47623233772285, 140.0019654200325, 140.31685261310218, 140.7008934199032, 140.73764915547463, 141.47857504067568, 141.79783607898793, 142.2268253113294, 142.4338700260237, 142.98255202451594, 143.30274961189664, 143.73343297342473, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $cu, 1$
  - q: 4.701080

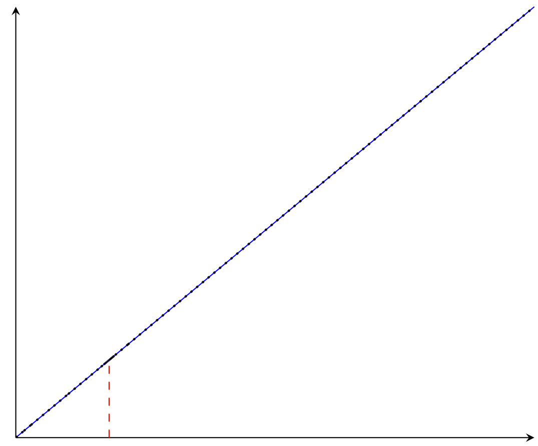
  - $\hat{q}^\beta$: 4.701080
  - q^β: 4.701080
  - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, 4.75409580038526, 4.8073435862836025, 4.813467676447643, 4.825601156120121, 4.844843256177314, 4.849719301830286, 4.882089573558703, 4.888121952381175, 4.9192700692457025, 4.956260392645724, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, 4.993351789275153, 5.030173826900646, 5.0359974771272995, 5.0418124100090695, 5.066729567155887, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

- $cu, 2$
  - q: 37.907411
  - $\hat{q}^\beta$: 37.907411
  - q^β: 37.907411
  - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65308365461216, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.68654427850372, 37.711317605988725, 37.71812813120871, 37.738762662315665, 37.7631578546152, 37.76995088018704, 37.78573239590169, 37.791030068986316, 37.81486687126635, 37.821651654621526, 37.84304537613103, 37.87287854392874, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

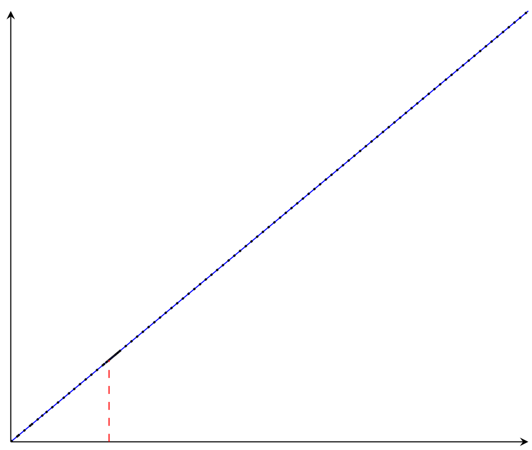
- $hu, 1$
  - q: 13.341988
  - $\hat{q}^\beta$: 13.341988
  - q^β: 13.341988
  - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.21041298113453, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.224123753786348, 13.22717920580809, 13.23798359102618, 13.238585544475612, 13.244985069124171, 13.251789114535772, 13.265751604070672, 13.266498143734903, 13.269097009439808, 13.279580305153486, 13.293477718720517, 13.294574825457861, 13.307344891293862, 13.30871858348944, 13.321736298873567, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000



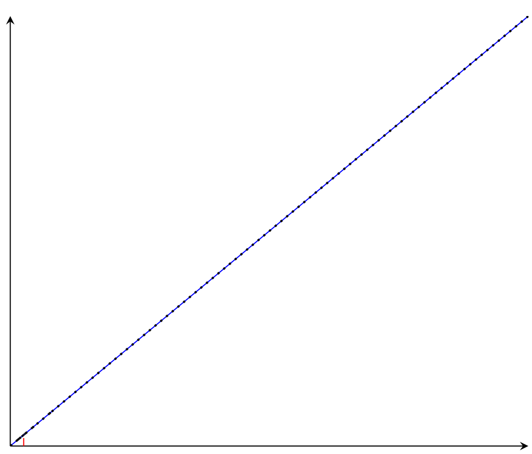
(a) q^β - $(1, 1, 1)$



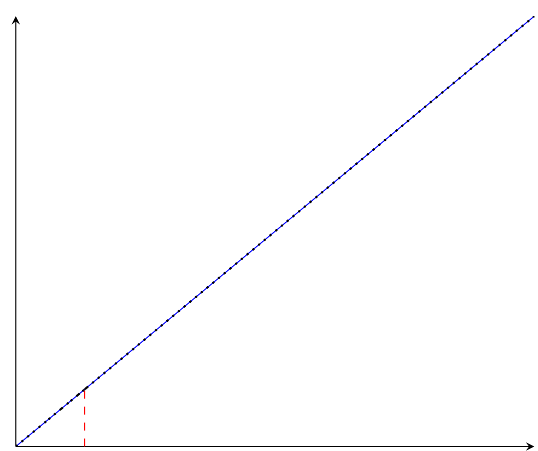
(b) q^β - $(1, 2, 2)$



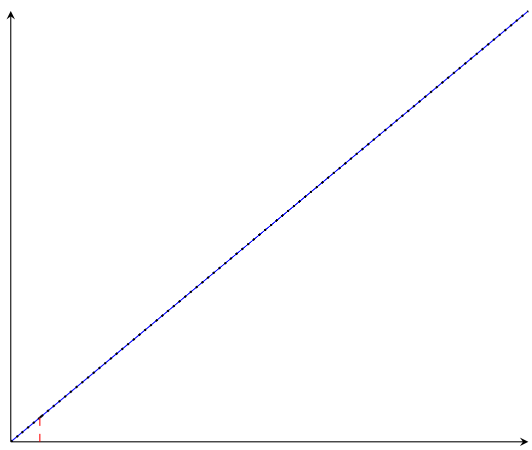
(c) q^β - $(2, 1, 2)$



(a) q^β - $cu, 1$



(b) q^β - $cu, 2$



(a) q^β - $hu, 1$

7 Other variables

Variable	Value
$z_{1,1,2}$	-0.0
$z_{1,2,1}$	-0.0
$z_{2,1,1}$	-0.0
$z_{2,2,1}$	-0.0
$z_{2,2,2}$	-0.0
$z_{hu,2}$	-0.0
$A^{\beta}_{1,1,2}$	0.0
$A^{\beta}_{1,2,1}$	0.0
$A^{\beta}_{2,1,1}$	0.0
$A^{\beta}_{2,2,1}$	0.0
$A^{\beta}_{2,2,2}$	0.0
$A^{\beta}_{hu,2}$	0.0
$A_{1,1,2}$	0.0
$A_{1,2,1}$	0.0
$A_{2,1,1}$	0.0
$A_{2,2,1}$	0.0
$A_{2,2,2}$	0.0
$A_{hu,2}$	0.0
$dt_{1,1,3}$	10.0
$dt_{1,2,1}$	280.0
$dt_{2,1,1}$	24.166666666666668
$dt_{2,2,1}$	220.0
$dt_{2,2,2}$	220.0
$dt_{2,2,3}$	240.0
$dt_{hu,2}$	180.0
$f^{H}_{1,1,2}$	0.0
$f^{H}_{1,2,1}$	7.538377520176221
$f^{H}_{2,1,1}$	20.0
$f^{H}_{2,2,1}$	0.0
$f^{H}_{2,2,2}$	9.017642916027217
$f^{C}_{1,1,2}$	4.792283781230658
$f^{C}_{1,2,1}$	0.0
$f^{C}_{2,1,1}$	0.5060418316959137

Variable	Value
$f^{C}_{2,2,1}$	13.0
$f^{C}_{2,2,2}$	0.0
$RecLMTD_{1,1,2}$	0.1
$RecLMTD_{1,2,1}$	0.01607047381606835
$RecLMTD_{2,1,1}$	0.1
$RecLMTD_{2,2,1}$	0.02011172883803818
$RecLMTD_{2,2,2}$	0.004340277777777596
$RecLMTD_{hu,2}$	0.005549329826813255
$q_{1,1,2}$	0.0
$q_{1,2,1}$	0.0
$q_{2,1,1}$	0.0
$q_{2,2,1}$	0.0
$q_{2,2,2}$	0.0
$q_{hu,2}$	0.0
$t^{H}_{1,1,2}$	370.91138383622507
$t^{H}_{1,2,1}$	650.0
$t^{H}_{2,1,1}$	590.0
$t^{H}_{2,2,1}$	590.0
$t^{H}_{2,2,2}$	590.0
$t^{C}_{1,1,2}$	410.0
$t^{C}_{1,2,1}$	500.0
$t^{C}_{2,1,1}$	568.662051917069
$t^{C}_{2,2,1}$	500.0
$t^{C}_{2,2,2}$	440.01120358475225
$b^{H,\text{in}}_{1,1,2}$	0.0
$b^{H,\text{in}}_{1,2,1}$	4899.945388114544
$b^{H,\text{in}}_{2,1,1}$	11800.0
$b^{H,\text{in}}_{2,2,1}$	0.0
$b^{H,\text{in}}_{2,2,2}$	5320.409320456058
$b^{H,\text{out}}_{1,1,2}$	0.0
$b^{H,\text{out}}_{1,2,1}$	4899.945388114544
$b^{H,\text{out}}_{2,1,1}$	11800.0
$b^{H,\text{out}}_{2,2,1}$	0.0

Variable	Value
$b^{H,\text{out}}_{2,2,2}$	5320.409320456058
$b^{C,\text{in}}_{1,1,2}$	1964.8363503045698
$b^{C,\text{in}}_{1,2,1}$	0.0
$b^{C,\text{in}}_{2,1,1}$	288.91391756876885
$b^{C,\text{in}}_{2,2,1}$	6500.0
$b^{C,\text{in}}_{2,2,2}$	0.0
$b^{C,\text{out}}_{1,1,2}$	1964.8363503045698
$b^{C,\text{out}}_{1,2,1}$	0.0
$b^{C,\text{out}}_{2,1,1}$	288.91391756876885
$b^{C,\text{out}}_{2,2,1}$	6500.0
$b^{C,\text{out}}_{2,2,2}$	0.0